

MUHAMAD KURNIAWAN FAUZI

Backend Engineer | DevOps & System Integration
Semarang, Indonesia (GMT+7) | +62 858-7905-6893 | krwnfauzi19@gmail.com
[linkedin.com/in/mkfauz](https://www.linkedin.com/in/mkfauz) | github.com/Mousynano | Remote-ready

SUMMARY

Backend Engineer with 11+ months of production experience building and operating FACETRO, a face-recognition attendance platform for presence tracking, reporting, and doorlock workflows. Delivered backend modularization, GCP-to-Huawei Atlas 500 edge AI migration, GitHub Actions + Watchtower deployment automation, Google SSO, MinIO object storage, REST APIs, and reporting workflows. Remote-ready for backend, DevOps, and system integration roles.

SKILLS

Programming: JavaScript/TypeScript, Node.js, Python, C/C++

Backend & APIs: REST API design, authentication, Google SSO, WebSocket, service modularization, admin systems, reporting automation

System Integration: service-oriented architecture, backend modularization, ML inference integration, role-based access workflows, operational reporting, edge AI deployment

Infrastructure & DevOps: Docker, Docker Compose, NGINX, GitHub Actions, Watchtower, Linux server administration, Git/GitHub

Databases & Storage: PostgreSQL, MySQL (schema design & queries), Prisma ORM, MinIO / S3-compatible object storage

Applied Research & IoT: ESP32/WSN, fingerprint/NFC attendance, metaheuristic/PID optimization, statistical testing

Languages: Indonesian (Native), English (Intermediate / professional working proficiency)

WORK EXPERIENCE

Backend Engineer Intern | PT Global Data Inspirasi (DataIns)

Sep 2024 – Jul 2025

- Built and operated FACETRO, a production face-recognition attendance platform supporting presence tracking, reporting, dashboard workflows, and doorlock-related operations.
- Implemented backend modularization from a monolithic deployment into separated services for core API, Telegram integration, object storage, and ML inference to improve maintainability and deployment flexibility.
- Supported recognition pipeline validation using Dlib and FaceNet, with reported average total processing time of 407.21 ms across backend and ML stages in phase testing.
- Migrated backend and ML services from a GCP-based stack to Huawei Atlas 500 on-premise edge AI hardware, including Docker image rebuilds for the target deployment environment.
- Implemented deployment automation using GitHub Actions for image builds and Watchtower with custom Docker Compose scripts for server-side redeployment.
- Integrated Google SSO, MinIO object storage/versioning, Excel/CSV reporting, dashboard data, and admin tooling for Presence V2, Doorlock V2, log edit/delete, and activity recap workflows.
- Developed REST API and operational workflows for user profiles, attendance logs, device logs, doorlock authorization, role-based Telegram broadcasting, and protected file access.

SELECTED PROJECTS

KKN Desa Gandu Website

2025

- Built and deployed a public village website using Next.js 15, Netlify, and Cloudflare DNS to support community documentation, local promotion, and digital archiving.

Applied Research & Embedded Systems

2024 – Present

- Built an ESP32-based IoT/WSN monitoring prototype for smart broiler farming, supporting real-time temperature and humidity data collection and wireless sensor network performance evaluation (Lecturer Research 2024).
- Developed Adaptive Cruise Control PID optimization research using GA, PSO, KMA, SKA, and HOA, evaluated through stability, robustness, time-response, and TTC safety metrics (International Collaboration of Top 200 Universities Research).
- Contributed manuscript writing and reproducibility preparation for a Deep Q-Network collision-avoidance study (Applied Science DIPA Research 2026).
- Built ESP32 firmware for a standalone fingerprint and NFC attendance device covering credential processing, presence logging, local configuration, and device control (UNNES X STEKOM Semarang Business Project).

EDUCATION

Bachelor of Computer Engineering | Universitas Negeri Semarang

2022 – Mar 2026

- GPA: 3.72 / 4.00 | Thesis: Adaptive Cruise Control optimization using metaheuristic algorithms and safety-oriented evaluation.

CERTIFICATIONS & COURSES

- Docker Foundations Professional Certificate — Docker, Inc.
- Associate Network Administrator — Digital Talent Scholarship
- Code Generation and Optimization Using IBM Granite — IBM
- Foundations of Project Management — Google LLC
- JavaScript Back-End Fundamentals — Dicoding Indonesia
- Career Essentials in Data Analysis – Microsoft & LinkedIn Learning
- Artificial Intelligence – Digital Talent Scholarship
- HCIA AI V3.5 Course – Huawei